Module 3 Report Addendum — Target Triage with Protein Language Models

Author: Krupa Patel | **Date:** Added October 28, 2026 in response to the October 25 rubric-check announcement.
Note: The original report (assessment report.pdf) remains unchanged.

A. Leakage Guard

Two panel proteins (SNCA and SH3GL2) were removed from the training panel prior to model fitting to prevent data leakage.

B. Agreement Between Prior and Module 2 Evidence

The Spearman rank correlation between the Module 2 multi-evidence scores and the best-modality tractability prior across my top 15 targets is **ρ = −0.618**. This substantial negative relationship demonstrates that disease relevance and sequence-based drug tractability are inversely aligned in this set. Disease-associated proteins with the strongest multi-evidence support (such as LMNA, PBXIP1, and SNCA) look the least "drug-like" based on sequence similarity to historically targeted proteins. Rather than indicating an error, this strong inverse relationship highlights that high-confidence genetic and transcriptomic disease targets often possess unique, hard-to-drug architectures or lack close sequence homologs among current clinical drug targets.

C. Justifying the Blending Weight (w = 0.6)

To integrate disease evidence with tractability, I selected a weighting of **w = 0.6** (60% Module 2 score, 40% tractability prior rank). This choice is based on three key analytical observations:

- **Input Reliability:** While the tractability prior demonstrates solid family-split validation performance (AUROC 0.754, AP 0.555), Module 2 multi-omics scores carry known statistical limitations—specifically, no individual gene passed false discovery rate (FDR) thresholding in the RNA layer. Because both inputs have inherent weaknesses, an even split (w = 0.5) serves as a baseline, but I tilted slightly toward Module 2 (w = 0.6) to ensure that disease association remains the primary driver of therapeutic triage.
- **Systematic Blind Spots of the Prior:** The sequence prior cannot infer tractability for targets lacking close sequence relatives in clinical databases. SNCA (α-synuclein) provides a clear demonstration: although SNCA is labeled as clinically precedented in the training panel (ab_clinical = True, given active monoclonal antibody trials), removing it during leakage guard resulted in a predicted antibody prior of p_ab = 0.001. Its nearest drugged panel neighbors (PSMD4, TNNT3, RPL7A; cosine similarities 0.84–0.85) represent the weakest matches among all 15 targets. Without sequence homologs in the panel, a target appears untractable even when clinically validated.
- **Sensitivity Across Weight Values:** Weight loop sensitivity testing (w = 0.4 to 0.8) proves that the top two targets, MAP4K4 and PBXIP1, remain entirely robust across the full range (swapping positions only at w = 0.8). The exact value of w primarily dictates the third-ranked pick: SNCA holds rank 3 at w = 0.5 and w = 0.6, whereas w = 0.4 favors FYN and w = 0.7 favors LMNA. Settling at w = 0.6 preserves SNCA in the top three while giving appropriate precedence to disease support.

D. Model Validation and Performance Gap

Evaluating the tractability prior yielded a random-split AUROC of 0.848 (AP 0.731) versus a family-split AUROC of 0.754 (AP 0.555; chance level 0.334). This performance gap illustrates the difficulty of generalizing across entire protein families compared to memorizing sequence patterns within random splits. To clarify previous phrasing: the family-split cross-validation holds out distinct *protein families* (GroupKFold by family_group), not structural folds, and the uncalculated metric "±0.02 fold variability" has been removed.

E. Interpretation of Anchor Variants

Variant scoring follows UniProt precursor numbering, where APOE C130R corresponds to mature C112R (ε4 allele) and APOE R176C corresponds to mature R158C (ε2 allele).

In testing, the risk-associated ε4 variant scored +7.21 (100th percentile within APOE), whereas the protective ε2 variant scored −8.56 (11.8th percentile). The model ranks these variants opposite to their clinical effect in Alzheimer's disease. This occurs because protein language models evaluate evolutionary log-likelihoods rather than clinical pathogenicity. Evolutionary sequence analyses show that arginine at codons 112 and 158 represents the ancestral primate and mammalian state (Hanlon & Rubinsztein, 1995). Because UniProt reference sequence APOE3 carries Cys-112/Arg-158, the C130R mutation restores the ancestral arginine at codon 112 (ε4 state), which the model assigns a high likelihood score. Conversely, R176C replaces the highly conserved arginine at codon 158 with cysteine (ε2 state), which the model heavily penalizes as a loss of a conserved ancestral residue.

Additionally, the established Alzheimer's risk variant TREM2 R47H scored at only the 47.7th percentile (−5.16). Rather than reflecting changes in overall protein embeddings, variant likelihood scores from ESM-2 650M measure the evolutionary acceptability of amino acid substitutions at specific positions. Evolutionary alignment accepts histidine at position 47 as moderately tolerable across species, masking the fact that R47H disrupts a critical basic surface patch required for ligand binding (Sudom et al., 2018).

F. Fact Corrections

- SNCA's predicted antibody tractability prior is p_ab = 0.001 (the original report erroneously listed 0.01).
- The sentence referencing "percentile-noise effects" in Section 1 of the original report is retracted.

G. Week 4 Target Selection

Although the combined ranking recommended MAP4K4 and PBXIP1, Module 4 analyzed MAP4K4 and SNCA. PBXIP1 was replaced with SNCA to evaluate a target central to Parkinson's pathology, despite the acknowledged trade-off that SNCA's high intrinsic disorder makes pocket identification and structure-based drug design exceptionally challenging compared to standard globular targets.

References

1. Hanlon, C. S., & Rubinsztein, D. C. (1995). Arginine residues at codons 112 and 158 in the apolipoprotein E gene correspond to the ancestral state in humans. *Atherosclerosis*, 112(1), 85–90.
2. Sudom, D., et al. (2018). Structural basis for mutated TREM2-mediated neurodegenerative disease risk. *Journal of Biological Chemistry*, 293(32), 12634–12646.